scientific reports

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A neuro-fuzzy multi-topology adaptive routing framework for QoS-aware healthcare IoT communications

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Healthcare IoT network loss their potential of reliability transmission, continuous monitoring, and intervention system due to critical network conditions along with its traffic characteristics. Thereby, it can directly affect the performance of the remote healthcare applications includes healthcare life quality check, patient safety, and clinical effectiveness in terms of QoS and robustness performance. In this paper presents a novel neuro-fuzzy multi-topology adaptive routing (NF-MTAR) method for enhancing reliable transmission of the healthcare IoT network. It integrates the neuro-fuzzy intelligence with multi-topology virtual partitioning enables dynamic optimization of the network resources based on network condition and its traffic characteristic. NF-MTAR method incorporates two unique innovations such as (i) neuro-fuzzy search engine identifies an optimum path to reach specific root node selected by incorporating five key parameters includes traffic flow intensity, resource utilization, residual energy, link quality, and node connectivity. (ii) Virtual Software Defined Networking (V-SDN) provides multi-topology virtual partitioning (elliptical, linear, and random) within the network, carry data transmission over multiple topologies for different traffic critical simultaneously. COOJA simulator is used to create three-layer 6LoWPAN architecture which capable of allowing dynamic network configuration and improve centralized policy management. Evaluation metrics are confirmed that the reasonable improvement is achieved such as high throughput (94.3%), reduce end to end delay (18.4ms), improve energy efficiency (31.2%), network lifetime (42.8%), and reliability (99.8%) by the proposed NF-MTAR method as compared to other state-of-art-methods. Thus, it provides potential improvement for next-generation medical monitoring and intervention system.

Keywords Virtual software defined network, Neuro-fuzzy search engine, Multi-topology virtual portioning, Healthcare IoT communication, Traffic characteristic and network condition

Healthcare monitoring system has been redefined by allowing patients to be closely monitored, consulting advice to be provided in real time and personal treatment options to be used. Internet of Things (IoT) platform supporting those wider applications within healthcare IoT ecosystems, several medical devices, sensors and different ways of communicating are used to collect, analyze and send vital information about patients onto complex networks[1,2]. Still, because healthcare data covers emergencies requiring instant response to everyday monitoring and office data, traditional networks find it challenging to ensure the right Quality of Service and reliable delivery needed for both patient and medical staff safety.

Typically, in healthcare IoT, IPv6 over 6LoWPAN routing protocols come with fixed root node choices and equal traffic handling which do not address the changing and many requirements of modern medical monitoring. These common methods have problems such as prioritizing urgent medical data poorly, not adjusting resources promptly as network variables vary and lacking response to changing patient and clinical workload on the network[3]. Poor network quality can result in delayed emergency help, faulty monitoring of vital signs and unstable methods for the clinic to make medical judgments which may risk patient safety. By adopting Software-Defined Networking (SDN) paradigms, healthcare IoT problems such as network management, enforcing rules and resource optimization, can be addressed more easily[4,5]. However, the IoT solutions based on SDN do not

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have the necessary intelligence to address the difficult optimization issues that health applications face which must evaluate minimizing latency, improving throughput, making the network energy efficient and ensuring reliability. Also, today's adaptive routing systems react to issues after they happen instead of predicting and stopping such issues when needed for critical healthcare services.

Recent studies clearly illustrated research gaps are present and limit the effectiveness of the existing IoT-based healthcare networking. First, the current adaptive routing protocols are normally single objective in terms of optimization, and they tend to favor either energy efficiency or latency minimization but do not offer the multi-dimensional performance requirements of healthcare applications[6]. Secondly, intelligent routing methods are frequently based on centralized machine learning models, which impose severe computational and possible single points of failure, rendering them unsuitable to healthcare IoT resource-constrained settings[7]. Third, existing multi-topology routing systems do not have the dynamic reconfiguration of their routing mechanisms needed to support the dynamism of healthcare processes and patient conditions. Lastly, current IoT network QoS architectures are not detailed enough to support medical services, wherein dissimilar medical data demands radically diverse service level assurances[8].

Neuro-fuzzy intelligence together with multi-topology adaptive routing is a new technology, which can possibly fill these research gaps by offering intelligent decision-making possibilities to fit in more dynamic network conditions and retain the level of transparency and interpretability needed in healthcare applications[9]. The integration of the learning capabilities of neural networks with the reasoning capabilities of fuzzy logic systems can allow neuro-fuzzy methods to offer strong performance with uncertain conditions and thus be able to continuously adjust to changing network patterns. This intelligent framework applied to multi-topology network architecture allows dynamically optimizing network partitioning of the virtual network depending on real-time traffic features and QoS needs, which can potentially offer substantial increases in the performance of healthcare IoT networks[10–12].

Research motivation and significance

IoT based healthcare applications require reliable data transmission for real time continuous monitoring is essential and it completely relay on IoT technologies. For that purpose, ultra-low latency network is normally considered for collecting the patient data remotely and it would be properly segregated, and it stored in the respective healthcare monitoring database. Such network follows certain defined network connections, where the edge router has permanent connection with root nodes to receive the patient data instantaneously. Due to this, repetitive data and different patient data are overwritten on the same patent field. Thereby, performance degradation may occur if this information is incorporated with intervention system (refer: Fig. 1a). Therefore, to avoid that, advance ultra-low latency network say Extended 6LoWPAN architecture provides more than one edge router is associated to fulfil the specific data acquisition from its fixed root nodes. The problem is to reach the root node of corresponding edge router is difficult under loss of root node connection due to the dynamic network conditions as well as varying traffic characteristic (refer: Fig. 1b). As a result, the corresponding edge router cannot receive the data and wastage of resource availability to compensate for the performance degradation towards impacting patient safety and clinical effectiveness. Therefore, to sort out this problem, dynamic root node selection can be done based on traffic characteristics and virtual portioning of network using software defined network.

Research objectives and contributions

Therefore, in this paper introduced a novel Neuro-Fuzzy Multi-Topology Adaptive Routing (NF-MTAR) framework mainly concentrated on providing reliable communication under dynamic network condition and traffic characteristics. In this NF-MTAR framework contains two core innovation such as.

(i) *Virtual software-defined network (V-SDN)* It provides multi-topology virtual partitioning induces different topologies such as elliptical, linear, and random. This is achieved by computing correlation metrics between five key parameters: traffic flow intensity, resource utilization, residual energy, link quality, and node connectivity. Based on the correlated outcome, the multi-topology virtual partitioning operation is activated by the proposed V-SDN method. Moreover, each topology is responsible for certain operations and is dynamically selected for data transmission. The responsibilities of the topologies are as follows:

- *Elliptical topology* is responsible for critical activities including emergency alerts and monitoring essential services, which have high-priority policies with high bandwidth allocation.
- *Linear topology* is responsible for non-critical activities such as routine health status updates and continuous background backups.
- *Random topology* is responsible for regular activities including flexible re-routing and bandwidth adjustment.

(ii) *Neuro-Fuzzy Search Engine* This plays a crucial role in dynamically adjusting the optimum path and selecting a suitable root node corresponding to the selected path. Different topologies are virtually formed, where each node participates in the multi-topology, and its individual weight is calculated. Based on the node weight, the method coordinates the nodes within the topology and computes the rank of each topology, which helps to organize the topologies accordingly. In this way, all kinds of healthcare data are accumulated in the appropriate server through proper coordination between the edge routers. Moreover, incorporating multi-topology virtual partitioning support ensures optimum utilization of available resources for specific types of patient data and ensures fine-grained QoS control to achieve superior performance. The contribution of the proposed NF-MTAR framework is given as follows:

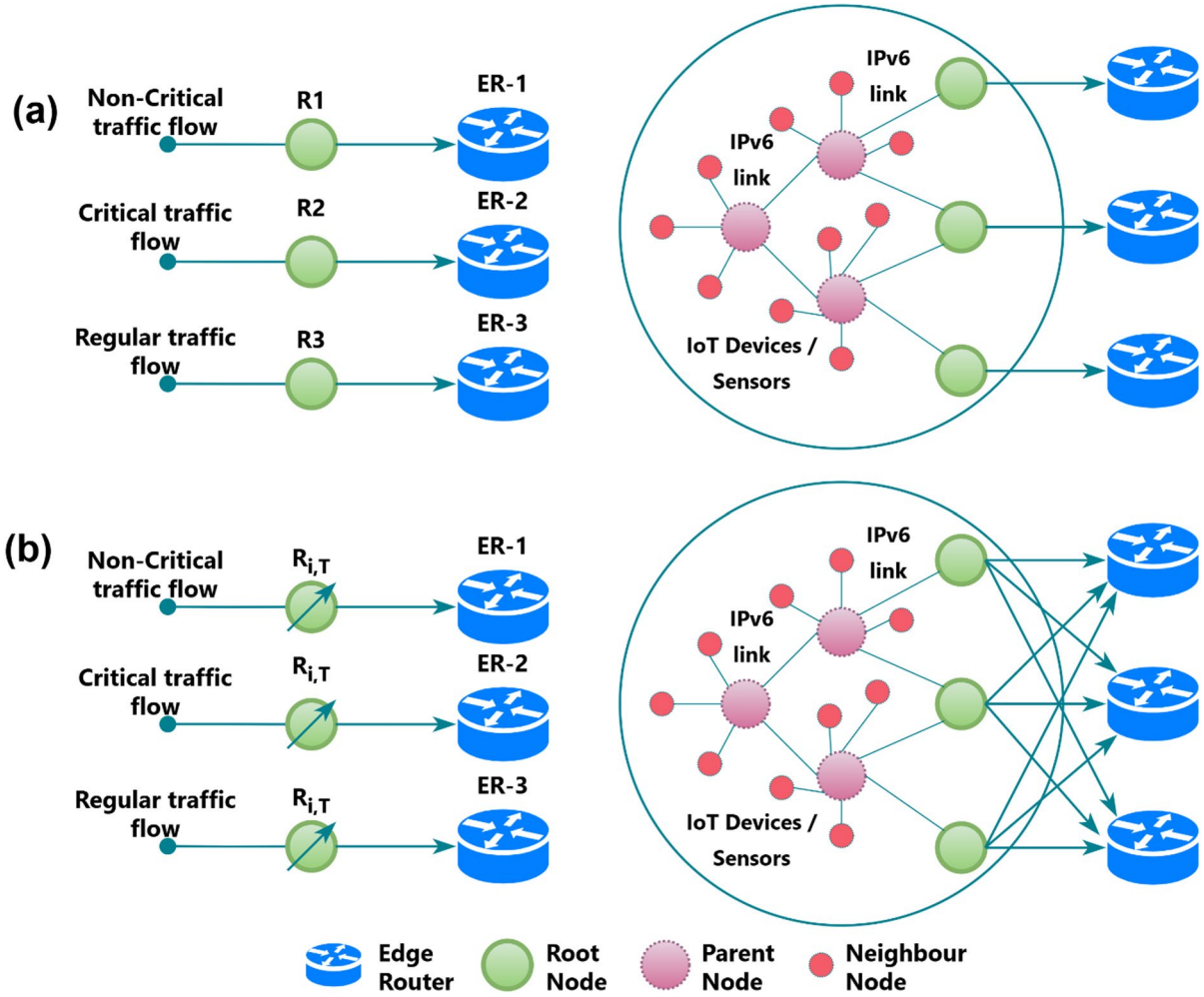


Fig. 1. illustrates the problem associated with advanced ultra-low latency networks. (a) a dedicated root node is fixed to an edge router for different traffic flows, and (b) dynamic selection of root nodes and edge routers for different traffic flows.

- We introduce a new virtual software-defined network approach that supports multi-topology virtual partitioning and ensures more effective data transmission carried out within this multi-topology environment. The approach dynamically coordinates multiple topologies based on rank computation using five key network parameters.
- For this virtual network partitioning, we have developed a unique mathematical model that clearly illustrates the selection of different topologies virtually based on traffic characteristics as well as dynamic network conditions.
- For proper evaluation, simulation is carried out by assigning specific QoS policy parameters under three distinct traffic characteristics including critical, regular, and non-critical traffic.
- The neuro-fuzzy search engine effectively utilizes its potential by incorporating a weighting mechanism to determine individual node capability as well as suitability for selecting the optimal network path.

The remainder of this paper is organized as follows: Sect. 2 gives the comprehensive review of related work in healthcare IoT networking, adaptive routing protocols, neuro-fuzzy systems, and multi-topology network. In Sect. 3, the design and details of the proposed multi-topology adaptive routing mechanism are explained. Section 4 includes results from thorough tests, along with comparison to other existing methods, investigates the lessons gained from the research, describes the present approach's shortcomings and outlines ideas for future improvement. Section 5 summary of key contributions and their significance for healthcare IoT networking.

Related works

In this section covers the detailed review of recent contribution of work towards Low-latency and high-efficiency signaling is possible as the digital networking landscape improves through new routing solutions, ways to optimize network traffic and advanced communication protocols.

Multi-topological routing in 6G

An open-source software-defined Wide Area Network (SD-WAN) is presented to handle communications between different clouds. Their work demonstrates that SD-WAN is key to linking all sorts of cloud services, resulting in networks that are easy to adapt, secure and reliable[13,14]. These researchers found that cloud networks play a crucial role in guiding traffic efficiently in high bandwidth and low latency circumstances. In event of real-time systems[15], offered a multi-topology routing method to enhance traffic with the applications of IEEE 802.1 Time-Sensitive Networking (TSN). Due to its flexibility to adapt to the new conditions of a network to enable Quality of Services (QoS), this method is significant to systems in industrial automation and vehicular networks because of time and data constraints. Moreover, emphasized on the creation of the routing protocols[16] through the design of timely routing style of networks based on MEO and LEO satellites. Therefore, their concepts of merging satellite communication with other diverse networks are quite appropriate to modern requirements of gaining internet access worldwide and supplement to intra-cloud communication plans.

In addition[17], discusses the variation in data center interconnectivity through introduction of FMPTCP that is aimed at utilizing the network resources in a more efficient way and minimizing the latency. This work shows that there is a need to have better routing protocols that enable data-driven applications to achieve their speed and data processing goals. Moreover[18], touch upon the issue of QoS through building a versatile adaptive multi-hop topology that is intended to meet the needs of a demand-sensitive data center. Through real-time studies of the traffic, they determine how topology in data centers must be changed, which can be an extension of the routing concepts proposed in the previous publications. This flexible nature of this type of network is reminiscent of the multi-agent reinforcement learning in dynamic routing[19]. Another technique has also emerged, and it is known as meta heuristics that integrates Takagi-Sugeno fuzzy systems with optimization to enhance adaptive routing within the wireless body area network[20,21]. The article contributes to the discussion of intelligent networks as it demonstrates that AI can have an impact on how routing is performed. Finally, concentrate on time-varying IoT networks with the help of Q-learning to enhance the performance indicators, which is close to the spirit of communication effectiveness optimization in large-scale distributed networks. Their paper highlights that there is a growing dependence on the use of machine learning to improve the ability to dynamically route in a wide range of networking conditions[22].

Network slicing approaches

IoT routing protocols are numerous studies that aim at making the transfer of data in 5G networks more efficient and flexible. Consequently, Adaptive Multipath Multi-Channel Energy Efficient (AMMEE) is also provided that gives priority to data rate and energy conservation, relevant in the implementation of IoT in 5G[23]. The comparison with QoS-AMMEE and QMR tests showed that their approaches are superior in terms of routing and resource management that make the network more reliable due to load- and bandwidth-aware patterns. Besides relying on routing mechanisms, taking advantage of reinforcement learning is key to increased network quality. As an example[24], came up with multi-agent techniques to restructure the topology of wireless mesh networks. It deals with the inconveniences associated with changeable network environments by providing flexible routing that can readily adjust to the changes. In addition[25], presents a model of routing and scheduling based on a multi-policy deep reinforcement learning strategy. Their methodology is basic when it comes to networks when multiple attributes of operation need to be targeted concurrently.

These observations make it apparent that AI-guided adaptive methodologies are a major factor in contemporary networking solutions. Intelligent networks and machine learning are also used to review the IoT energy harvesting systems that imply the identification of the most suitable path to choose to get the information in energy-saving environments[26–28]. They demonstrate that strategies in which numerous factors are treated as units can greatly enhance performance of networks in terms of addressing energy issues. Moreover[29], said network management is also manageable using controlled SDN overlays that increase the ductility of virtual LANs that result in converging tools that enhance the effectiveness of networks. Moreover, the ideas of graph reinforcement learning introduced in[30] are a foundation of a new approach to resource management in ad hoc networks. This new technique is the best way of routing and allocating radio resources because it addresses the numerous problems encountered by wireless networks nowadays[31]. This paper contributes to understanding the ways in which intelligent systems contribute to the flexible and successful management of networks under various conditions. Lastly, the researchers investigated how federated learning could be organized using blockchain in 5G to manage resources efficiently and sustain network slicing[32,33]. The paper also describes the reasons why the increasing requirements of ionic solutions are increasingly relying on networking which emphasizes efficiency, security and flexibility.

Materials and methodology

The proposed NF-MTAR framework provides a complete and structured solution to improve health monitoring systems with intelligent routing and management of networks. The architecture starts from the top with the SDN controller layer handling all the control and direction for network management. This layer controls and organizes the network functions by using main rules and flow settings, while also choosing the root in the network, organizing quality-of-service policies and classifying traffic to uphold efficient performance in all kinds of healthcare monitoring. Figure 2 shows the Proposed NF-MTRA framework architecture diagram.

Overview of proposed NF-MTAR

The neuro-fuzzy multi-topology decision engine layer plays a central role including computer ranks and builds the objective function which allows it to react intelligently to recent changes on the network. After evaluating traffic load, the engine sets its route, manages key nodes and looks for ways to keep itself running smoothly and efficiently. Such a system can improve accuracy in reading and update by itself even under the unpredictable

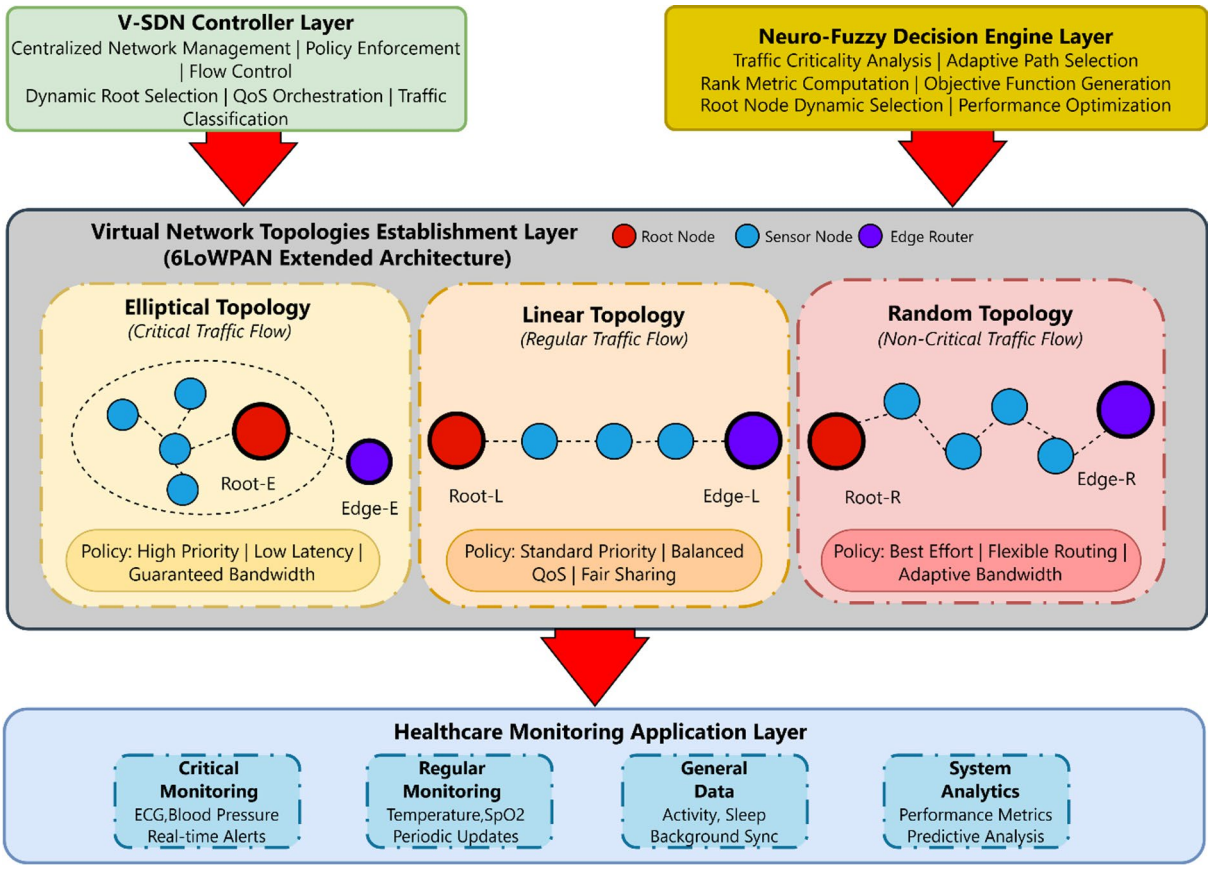


Fig. 2. Proposed NF-MTRA framework architecture diagram.

changes in the network. This layer makes the extended 6LoWPAN architecture workable, by forming three different virtual networks to manage healthcare dataindicating strong potential for future application in real medical settings upon hardware implementation and clinical validation. Elliptical Topology is created to support critical activities like emergency alerts and monitoring essential services, providing these with high-priority policies and guaranteed speed and allocation of bandwidth. Regular checks on a patient's vital signs and routine updates about their health are managed through Linear Topology by assuring all traffic receives equal attention. Random Topology is an appropriate choice where the non-vital traffic like background backup and general monitoring is concerned, best-effort progress is used, and the bandwidth can be adjusted and rerouted as the demands vary. Each of the topologies has a method of picking the dynamic root node, which makes the network more workable as per the occurrences within the network and the traffic. NF-MTAR framework suitable for specific real time medical monitoring environment as shown in the healthcare application layer. That implies, this framework has potential of supporting tools of critical chronic diseases such as ECG and blood pressure monitoring and those of regular readings such as temperature or SpO2 levels. Experience layer consists of general data gathering for activity, sleep monitoring and background syncing, as well as analytics for tracking the system's performance and making health estimates. Combining application demands with selecting the right network topology guarantees every kind of healthcare data gets the appropriate network support it needs according to its criticality and timeliness.

Virtual network partitioning establishment

Virtual Software Defined Network (V-SDN) provide controller layer establishes the foundational framework for centralized network management and virtual network partitioning in the extended 6LoWPAN architecture. The network partitioning function is defined in Eq. (1) which helps divide the entire network into three distinct virtual topologies sorted based on traffic characteristics. As a result, key health data, updates from regular monitoring and non-critical background data can access the specialized network structures optimized for their level of Quality of Service (QoS) requirements.

$$V = V_E \cup V_L \cup V_R \tag{1}$$

where:

$V_E = \{v_1, v_2, ..., v_{ne}\}$ represents nodes in elliptical topology for critical traffic.
$V_L = \{v_1, v_2, ..., v_{nl}\}$ represents nodes in linear topology for regular traffic.
$V_R = \{v_1, v_2, ..., v_{nr}\}$ represents nodes in random topology for non-critical traffic.

$V_E \cap V_L \cap V_R = \varphi$ (mutually exclusive partitions).

Equation (2) determines best position of identifying virtual topologies guided by rank metric computed through topology assignment using maximum likelihood estimation.

$$T(v_i) = argmax P_E(v_i), P_L(v_i), P_R(v_i) \tag{2}$$

where $P_t(v_i)$ indicates probability occurrence of topology v_i assigned belonging to topology $t \in (E, L, R)$ based on traffic criticality analysis. V-SDN follows a QoS specific policy enforcement matrix to configure suitable virtual topologies including elliptical, linear, and random dynamically selected with respect of traffic characteristics such as critical, regular, and non-critical, respectively. The configuration matrix is defined in Eq. (3) as:

$$\Pi = [\pi_{ij}]_{3\times m} = \begin{bmatrix} \pi_{E1} & \pi_{E2} & \cdots & \pi_{Em} \\ \pi_{L1} & \pi_{L2} & \cdots & \pi_{Lm} \\ \pi_{R1} & \pi_{R2} & \cdots & \pi_{Rm} \end{bmatrix} \begin{matrix} \text{(Elliptical-Critical)} \\ \text{(Linear-Regular)} \\ \text{(Random-Non-critical)} \end{matrix} \tag{3}$$

where π_{ij} represents the policy parameter j for topology i, and m is the number of policy parameters (bandwidth allocation, latency constraints, priority levels, etc.).Each row in the matrix contains $m = 12$ QoS-specific policy parameters, listed in Table 1.

Since the matrix structure is used, the SDN controller can closely manage network resources such as bandwidth, latency rules, priority levels and security settings for every topology type. The QoS constraint enforcement expression in Eq. (8) controls all policies so that critical healthcare systems are given sufficient access to network resources which allows the network to maintain maximum performance. Using an indicator function, this equation enables decision-making that is always precise according to whether each node belongs to the cluster or not, enabling precise resource allocation based on node topology membership.QoS enforcement for a virtual node v_i under topology $t \in E, L, R$ is governed by a weighted sum of its policy parameters:

$$\mathrm{QoS}_t(v_i) = \sum_{j=1}^{m} \omega_j \cdot \pi_{tj} \cdot \delta(v_i \in V_t) \tag{4}$$

where: ω_j is the weight factor for policy parameter j, $\delta(v_i \in V_t)$ is the indicator function (1 if node v_i belongs to topology t, 0 otherwise). Where, the priority weights ω_j are assigned based on QoS sensitivity:

$$\omega_j = \begin{cases} 0.25, & \text{for } j \in \{1,2,3,4\}\text{(High Priority)} \\ 0.15, & \text{for } j \in \{5,6,7,8\}\text{(Medium Priority)} \\ 0.05, & \text{for } j \in \{9,10,11,12\}\text{ (Low Priority)} \end{cases} \tag{5}$$

The core task of the SDN controller can be seen in Eq. (6) by the global network control objective function which minimizes delay ($D_t(v_i)$), packet loss ($L_t(v_i)$and resource utilization ($U_t(v_i)$), across all the different created virtual topologies. Guaranteeing minimum quality of service for all topologies, the constraint set selects only one topology for each node.

$$min\Phi = \sum_{i=1}^{n} \sum_{t\in\{E,L,R\}} [\alpha_t \cdot D_t(v_i) + \beta_t \cdot L_t(v_i) + \gamma_t \cdot U_t(v_i)] \tag{6}$$

subject to:

Parameter	Description	Critical (π_{Ej})	Regular (π_{Lj})	Non-Critical (π_{Rj})
π_{i1}	Bandwidth Allocation (%)	80	15	5
π_{i2}	Priority Level (1–7)	7	4	1
π_{i3}	Maximum Latency Threshold (ms)	25	100	500
π_{i4}	Minimum Throughput Guarantee (%)	95	85	70
π_{i5}	Packet Loss Tolerance (%)	0.1	1	5
π_{i6}	Energy Efficiency Weight (0–1)	0.3	0.5	0.8
π_{i7}	Security Level	AES-256	AES-128	Basic
π_{i8}	Redundancy Factor (replicas)	3	2	1
π_{i9}	Buffer Size Allocation (KB)	512	256	128
π_{i10}	Retransmission Limit (count)	5	3	1
π_{i11}	Route Update Frequency (s)	1	5	30
π_{i12}	QoS Monitoring Interval (ms)	100	1000	10,000

Table 1. QoS-specific policy parameters.

- $\sum_{t\in\{E,L,R\}} x_{it} = 1 \forall\, i$ (each node assigned to exactly one topology).
- $\mathrm{QoS}_t(v_i) \geq \mathrm{QoS}_{min}^t \forall\, v_i \in V_t$ (minimum QoS guarantee).

Because of this, the SDN controller can determine the best network operations for all connected nodes as well as for the whole system as a whole. With the weight coefficients ($\alpha_t, \beta_t, \gamma_t$), the system can select which performance metrics are most important, depending on what is happening on the network and in the application. By using binary assignment variables (x_{it}), the model is mathematically accurate and still compatible with actual networks.Using the policy matrix, the SDN controller dynamically generates Open Flow-based flow rules for each node v_i in topology t as:

$$FR_t(v_i) = \left\{ \begin{array}{c} \text{match:}src = v_i, \\ \text{action:}forward = t, \\ \text{priority:}\pi_{t2}, \\ \text{queue:}\pi_{t1} \end{array} \right\} \tag{7}$$

NF-MTAR routing mechanism

Dynamic root node selection

Using both neuro-fuzzy search engines and multi-topology virtual portioning control, NF-MTAR routing finds the best routes and tends to automatically replace the root coordinator in networks with many connected topologies, sample scenarios to show routing mechanism as seen in Fig. 3. It applies to a neuro-fuzzy model to determine which root nodes are selected according to the network situation and kinds of traffic. In Eq. (6), the comprehensive network parameter vector contains the important network state information such as x_1: Traffic Flow Intensity $\rightarrow$ {Very Low, Low, Medium, High, Very High}, x_2: Resource Utilization $\rightarrow$ {Very Low, Low, Medium, High, Very High}, x_3: Energy Consumption $\rightarrow$ {Very Low, Low, Medium, High, Very High}, x_4: Link Quality $\rightarrow$ {Good, Poor}, and x_5: Node Connectivity $\rightarrow$ {Low, Medium, High}, respectively.

$$X(v_i, t) = \left[\begin{array}{c} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{array} \right] \tag{12}$$

Because the system has this representation, it can adapt its routing by considering the real network state, not only fixed data. The adjustable nature of the temporal component in the parameter vector is very helpful in healthcare monitoring environments where patterns of network traffic may fluctuate a lot depending on patients and their needs.

Neuro-fuzzy objective function generation

The proposed Neuro-Fuzzy Decision Engine incorporates an expert-derived rule base composed of $R = 125$ rules. It Configured through a $53 \times 2 \times 3 = 750$ input space and pruned to 125 valid and non-redundant rules using domain knowledge and correlation filtering.The intelligence behind the routing is mainly expressed by the neuro-fuzzy equations presented as Eq. (3) to (8). The joining of different fuzzy membership values, as guided by Eq. (9), makes the system able to address the uncertainty and imprecision present in measurement data.

$$\mu_{A_j}(x_i) = \exp\left(-\left(\frac{x_i - c_j}{\sigma_j} \right)^2 \right) \tag{9}$$

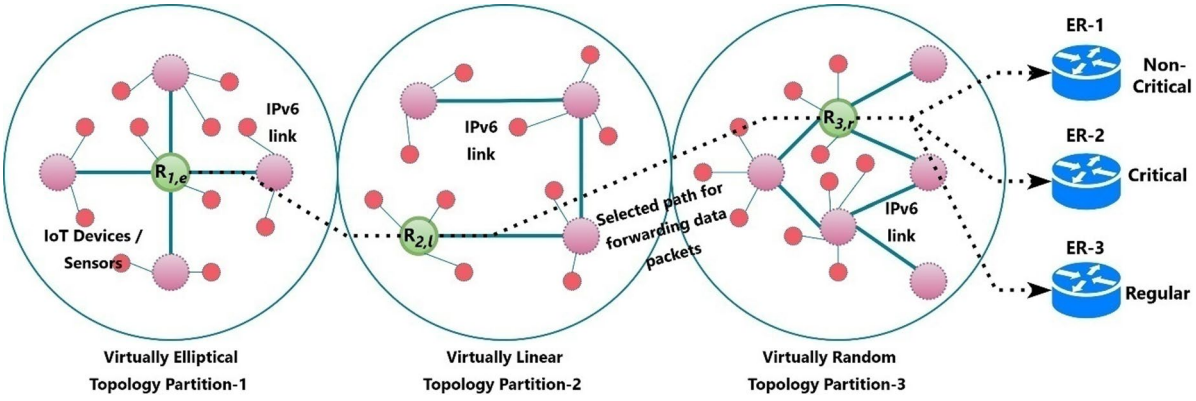


Fig. 3. illustrates the scenario of root node selection dynamically with respect to traffic load condition and resource constraints.

where $\mu_{A_j}(x_i)$ is the Gaussian membership function for the j-th fuzzy set of input x_i, with center c_j and width σ_j.

In Eq. (14), the expert rules are stored and in Eq. (13), the system judges the significance of each rule for the current state of the network.

$$R_k: \text{ IF } x_1 \text{ is } A_{1k} \text{ AND } x_2 \text{ is } A_{2k} \text{ AND } \ldots \text{ AND } x_5 \text{ is } A_{5k} \text{ THEN } y \text{ is } B_k \tag{10}$$

Examples of selected rules from the pruned rule base:

Rule 1:

$$\text{IF}(x_1 = \text{VeryHigh}) \wedge (x_2 = \text{High}) \wedge (x_3 = \text{Low}) \wedge (x_4 = \text{Good}) \wedge (x_5 = \text{High})$$
$$\Rightarrow \text{ Suitability: Excellent}, w = 0.95$$

Rule 47:

$$\text{IF}(x_1 = \text{High}) \wedge (x_2 = \text{Medium}) \wedge (x_3 = \text{Medium}) \wedge (x_4 = \text{Good}) \wedge (x_5 = \text{Medium})$$
$$\Rightarrow \text{ Suitability: Good}, w = 0.78$$

Rule 98:

$$\text{IF}(x_1 = \text{Low}) \wedge (x_2 = \text{VeryHigh}) \wedge (x_3 = \text{VeryHigh}) \wedge (x_4 = \text{Poor}) \wedge (x_5 = \text{Low})$$
$$\Rightarrow \text{ Suitability: Poor}, w = 0.15$$

where R_k represents the k-th fuzzy rule, and B_k is the consequent fuzzy set for output y.

$$w_k = \prod_{i=1}^{5} \mu_{A_{ik}}(x_i) \tag{11}$$

In Eq. (14), the defuzzification process ensures fuzzy reasoning results are real values that can be immediately used by routers.

$$f(X(v_i, t)) = \frac{\sum_{k=1}^{125} w_k \cdot y_k \cdot \mu_{B_k}(y_k)}{\sum_{k=1}^{125} w_k \cdot \mu_{B_k}(y_k)} \tag{12}$$

$$f_{\text{norm}}(X(v_i, t)) = \frac{f(X(v_i,t)) - f_{min}}{f_{max} - f_{min}}, f_{min} = 0.1, f_{max} = 1.0 \tag{13a}$$

$$\text{Confidence}(v_i, t) = \frac{\max_k(w_k)}{\sum_{k=1}^{125} w_k} \tag{13b}$$

Where R is the total number of fuzzy rules and y_k is the center of the consequent fuzzy set B_k. Equation (16a) and (16b) be the normalized put form and confidence measure, respectively.

The neural network learning component, defined through Eqs. (15) and (17), provides adaptive capability by continuously updating the system parameters based on network performance feedback, ensuring that the routing mechanism improves its decision-making accuracy over time.

$$\theta_j(t+1) = \theta_j(t) - \eta \cdot \frac{\partial E}{\partial \theta_j} \tag{17}$$

where: θ_j represents the j-th network parameter (weights and biases), η is the learning rate, and E is the error function defined as:

$$\eta(t) = \eta_0 \cdot exp(-\tfrac{t}{\tau}) \tag{18}$$

with $\eta_0 = 0.01, \tau = 500$.

$$E = \tfrac{1}{2N} \sum_{i=1}^{N} (y_{\text{desired}}(v_i) - f(X(v_i, t)))^2 + \tfrac{\lambda}{2} \sum_j \theta_j^2 \tag{19}$$

Rank metric computation

Equations (18) and (19) were used to develop the rank metric framework, creating a single scoring system for judging the suitability of nodes as roots in every virtual topology. By uniting the output of the neuro-fuzzy model with metrics such as hop distance and communication cost, this criterion helps to evaluate networks based on their reasoning power and main features. Because of the dynamic root selection, the best node is chosen as the root for every topology as needed, responding to shifts in network behavior and data flow. Because of adaptive weight coefficients, the system can give higher importance to different evaluation factors based on what is needed at the current moment and from the network.

$$R(v_i, t) = w_1 \cdot f(X(v_i, t)) + w_2 \cdot H(v_i, t) + w_3 \cdot C(v_i, t) \tag{18}$$

where: $f(X(v_i, t))$ is the neuro-fuzzy output score, $H(v_i, t)$ is the hop distance to edge router, $C(v_i, t)$ is the communication cost metric, and w_1, w_2, w_3 are adaptive weight coefficients.

$$r^*(t) = \arg \min_{v_i \in V_t} R(v_i, t) \tag{19}$$

where $r^*(t)$ is the optimal root node for topology t at time. The new model, introduced by Eqs. (20) and (21), uses multi-criteria optimization to choose a path based on distance, link condition and energy used along the path. Because of this detailed cost function, the routes selected still help with efficient data transport while maintaining the longevity and trustworthiness of the network.

$$C(P_{i \to j}) = \sum_{k=1}^{K-1} [\lambda_1 \cdot d(v_k, v_{k+1}) + \lambda_2 \cdot q(v_k, v_{k+1}) + \lambda_3 \cdot e(v_k)] \tag{20}$$

where: $P_{i \to j}$ represents the path from node v_i to edge router v_j, K is the number of hops in the path, $d(v_k, v_{k+1})$ is the distance metric between consecutive nodes, $q(v_k, v_{k+1})$ is the link quality metric, $e(v_k)$ is the energy cost at node v_k, and λ_1, λ_2, λ_3 are cost weighting factors.

$$P^*_{i \to j} = \arg \min_{P \in \Phi_{i \to j}} C(P_{i \to j}) \tag{21}$$

where $\Phi_{i \to j}$ is the set of all feasible paths from node v_i to edge router v_j.In path optimization, the system ensures the virtual network meets specific needs by giving the most suitable routing strategies to each traffic type. Using math, the algorithm can recognize the most efficient routes as the conditions on the network change, unlike the fixed topology.

Traffic flow analysis and topology adaptation

The traffic flow analysis and topology adaptation mechanism discussed in Eqs. (22) to (24) supports dynamic changes in a network determined by patterns of traffic. The system uses traffic flow characterization to determine the arrival speeds of each traffic type and notice big shifts in network demand.

$$TF(t) = \begin{bmatrix} \lambda_c(t) \\ \lambda_r(t) \\ \lambda_n(t) \end{bmatrix} \tag{22}$$

Where $\lambda_c(t)$, $\lambda_r(t)$, $\lambda_n(t)$ represent arrival rates for critical, regular, and non-critical traffic respectively. By applying a threshold-based analysis, the mechanism can tell if network changes are needed, so no extra adjustments are performed but significant traffic changes are quickly reacted to. A hysteresis element is introduced into the root node change decision to protect the network from negative effects of oscillations.

$$\Delta(t) = \| TF(t) - TF(t - \Delta t) \| > \tau_{threshold} \tag{25}$$

Here, $\Delta(t)$ is the monitoring interval and $\tau_{threshold}$ is the adaptation threshold.

$$\text{Change_Root} = \begin{cases} \text{True}, & \text{if } R(r_{\text{current}}, t) - R(r^*, t) > \epsilon \\ \text{False}, & \text{otherwise} \end{cases} \tag{24}$$

Where ε is the root switching threshold to prevent frequent oscillations.The NF-MTAR framework provides its highest level of performance without leaving out any of the required goals in every virtual topology. Such an approach means that healthcare monitoring systems have access to dependable, adaptable and efficient networks that strictly guarantee high standards.

Results and discussion

This section provides proper comprehensive evaluation of the proposed NF-MTAR framework against other state of art methods such as traditional RPL methods (Multi-topology TSN[3], Bandwidth aware N-channel[16], adaptive methods (Dynamic multi-hop[7], adaptive FANET[21], Graph RL routing[24], optimization methods (TSFIS-GWO[12], opt RISQL[14], multi-agent RL method (Fuzzy baseline[18], Intelligent energy harvesting[23], respectively. Evaluation mainly focused on (i) traffic type throughput analysis, (ii) traffic type latency analysis, (iii) temporal and scalability analysis and (iv) statistical validation. Each analysis is performed with more detailed aspects such as comparison with respect to traffic types, improvement profile, distribution analysis, and percentile analysis. These analyses provide strong evidence for performance superiority of the proposed NF-MTAR framework.

Simulation environment and parameters

The proposed NF-MTAR framework is studied by running computer simulations on the COOJA simulator which is part of the Contiki-NG operating system used for 6LoWPAN networks. In the simulation, we used a healthcare scenario with over 100–200 sensor nodes in a heterogeneous network that offered three different virtual formations (elliptical, linear and random) for different types of traffic. The parameters measured by the network were a 50-meter range, different packages sizes from 64 bytes (critical alert data) to 1024 bytes (regular monitoring) and up to 10 packets/second during critical care monitoring versus 1 packet/second for routine logs. The search engine was created using MATLAB Fuzzy Logic Toolbox and needed five input parameters(traffic load, available resources, energy, connectivity degree and past history). It was set up with 125 fuzzy rules taken

from expert recommendations within healthcare networking standards. By using back propagation learning with a learning rate of 0.01 and momentum factor of 0.9, the mechanism allowed the neural network to gradually make better routing choices over time as it captured performance data. The extended 6LoWPAN architecture used SDN, with the Open Daylight framework handling central rules for policy and dividing virtual networks properly.

Traffic-Type throughput analysis

Figure 4a depicts the throughput performance across different traffic types. From the results, it conveys that the proposed NF-MTAR framework achieved higher performance as compared with existing baseline methods in progressively. Such as critical traffic (95.2%), regular traffic (87.3%), and non-critical traffic (78.9%), respectively. this comparison brings solid confirmation about the proposed NF-MTAR framework consistently higher margin across under three distinct traffic types critical, regular, and non-critical in the following ways such as (i) it is highest among all baselines, (ii) significantly above competitors, and (iii) maintains superiority even for lowest priority, respectively. Especially, on comparing performance of two specific methods say traditional methods includes (Critical (71.4%), Regular (68.2%), and Non-Critical (62.1%)) and adaptive methods includes (Critical (76.8%), Regular (73.5%), and Non-Critical (67.4%)) under three distinct traffic types, respectively. Another important aspect to explore the significant improvement achieve by proposed NF-MTAR framework is defined through traffic differentiation effectiveness. According to this, NF-MTAR shows nearly 16.3% of performance spread between critical and non-critical traffic. Meanwhile, traditional RPL shows 9.3% of performance spread, which indicates poor QoS differentiation. Thus, confirms that the multi-topology virtual portioning approach effectively prioritizes for critical healthcare data transmission.

Figure 4b shows the comparative analysis of performance improvement of proposed NF-MTAR framework over different baseline methods under different traffic characteristic includes critical, regular, and non-critical. Under critical traffic scenarios, the key observation conveys that the reasonable percentage of improvement is achieved by the proposed NF-MTAR framework such as traditional RPL methods (33.4%), adaptive methods (24%), optimization methods (~18.4%), and multi-agent RL methods (~19.6%), respectively. Similarly, the proposed NF-MTAR framework shows consistency gains and moderate improvement range (15–20%) across all above baseline's methods under both regular and non-critical traffic, respectively. Therefore, it is confirmed that the proposed NF-MTAR framework achieves largest gain against traditional RPL methods and smaller with consistency gain against adaptive methods.

Figure 4c shows overall performance metrics. It clearly shows the peak performance (95.2%) and minimum performance (78.9%) of the proposed NF-MTAR framework. this conveys that under optimal condition the system is extremely exhibits peak performance and maintains minimum 80% throughput even under lowest priority traffic. Figure 4d illustrates throughput distribution analysis. NF-MTAR shows narrow, tall peak around 85–95% of the throughput range. It is highly concentrated near optimal values and right skewed indicates frequently occurring high performance state. Baselines methods show distributions as wider, flatter around 60–80% of the throughput range. However, these methods show more dispersed frequency confirming inconsistent

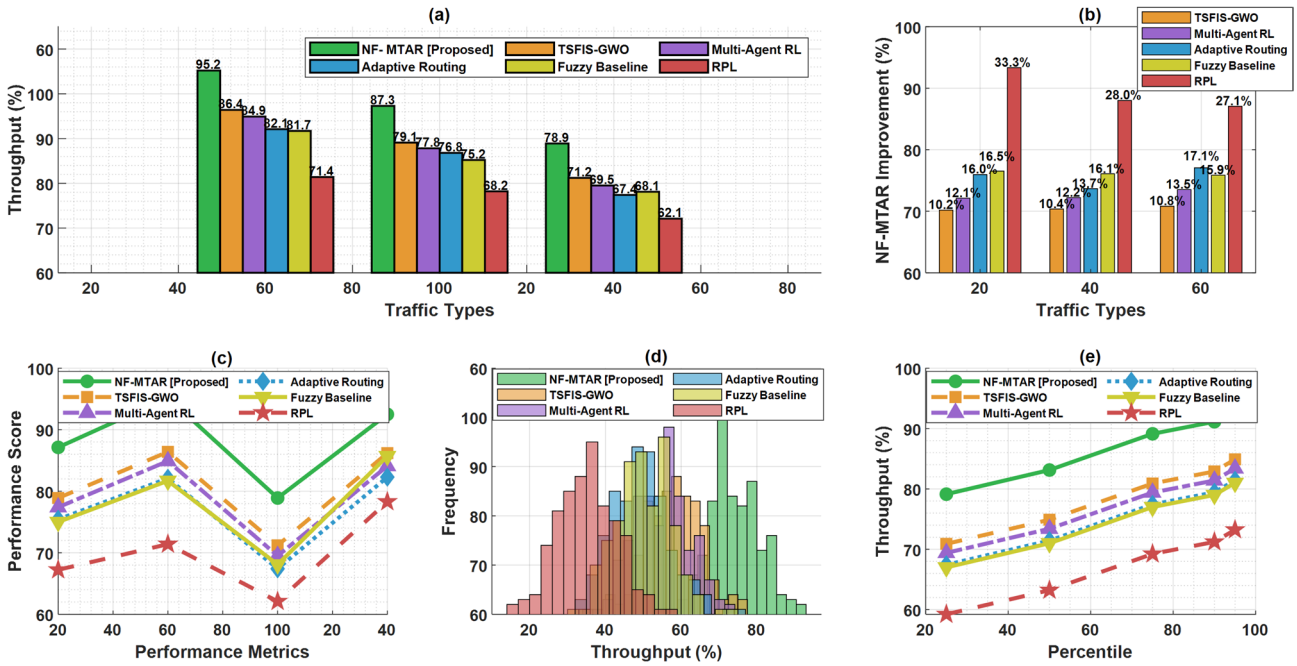


Fig. 4. Traffic-type throughput analysis. (**a**) Throughput performance by traffic type, (**b**) performance improvement over baselines, (**c**) overall performance metrics, (**d**) throughput distribution analysis, and (**e**) throughput percentile analysis.

performance. It is simply observed through bimodal patterns indicating unstable behaviour. The variance of the distribution spread conveys important inference which is gives as: NF-MTAR (80–95%), traditional methods (55–85%), optimization methods (60–85%) and adaptive methods (65–85%), respectively.

Figure 4e depicts throughput percentile analysis under 50th percentile, 75th percentile, 90th percentile, and 25th percentile. That implies, medium performance is achieved at 50th percentile for specific methods such as (~ 88%) of NF-MTAR [proposed], (~ 82%) of TSFIS-GWO[12], (~ 80%) of Multi-Agent RL[17], (~ 76%) of adaptive FANET[21], (~ 72%) of Fuzzy baseline[18], and (~ 68%) of RPL[16], respectively. These outcomes conveys that the proposed NF-MTAR framework quickly reaches maximum peak point performance and maintains it consistently. Other baseline methods are more gradual and linear progression. Therefore, overall observation forms the traffic type of throughput analysis conveys that the proposed NF-MTAR framework capable for consistent patient monitoring with high performance and reduces risk of missed critical alerts. Moreover, these analysis report more supportive for tight distribution allows accurate network planning and capacity estimation.

Traffic-type specific latency analysis

Figure 5a illustrates the end-to-end latency comparison with traffic types. It clearly inferred that the proposed NF-MTAR framework shows exceptional low latency performance with respect to three distinct traffic types such as critical traffic (18.4ms), regular traffic (45.2ms), and non-critical traffic (89.7ms), respectively. The significant improvement of the latency reduction of NF-MTAR can be understand by comparing with baseline methods especially traditional RPL under three distinct traffic types. Such as critical traffic (67.3ms), regular traffic (94.8ms), and non-critical traffic (156.2ms), which indicates 3.66× slower, 2.10× slower, and 1.74× slower than NF-MTAR, respectively. The critical traffic plays important role to affect the performance and thereby, comparing critical traffic analysis between other two baseline methods include adaptive methods and multi-agent RL method. It is clearly inferred that critical traffic: ~38.9ms (still 2.11× slower than NF-MTAR) for adaptive methods and thus shows 20.5ms difference gap with NF-MTAR. Sub sequentially, multi-agent RL method shows critical traffic: ~44.6ms (2.42× slower than NF-MTAR). Therefore, NF-MTAR critical alerts receive substantially better treatment relative to background traffic.

Figure 5b shows the latency reduction analysis with state of art methods. It is evidently inferred that the proposed NF-MTAR framework reduces latency to maximum extent as compared with other baseline methods. Such as (i) Traditional RPL: 72.6% reduction (67.3ms → 18.4ms), (ii) Fuzzy baseline: 55.4% reduction (~ 88.4ms → 18.4ms), (iii) Adaptive methods: 64.7% reduction (~ 83.5ms → 18.4ms), (iv) Optimization methods: 52.7% reduction (38.9ms → 18.4ms), and (v) Multi-Agent RL: 58.7% reduction (44.6ms → 18.4ms), respectively. therefore, overall performance improvement pattern of proposed NF-MTAR framework under critical traffic (72.6%), regular traffic (52.3%), and non-critical (42.6%), thus demonstrates robustness and superiority as compared with other baseline methods.

Figure 5c depicts latency distribution analysis. It is clearly noticed that the NF-MTAR framework shows distributed characteristic as narrow, sharp peak concentrate around (15-20ms) for critical traffic. More than (>90) occurrence lies on optimal range where high frequency exhibits minimum variance and shows sharp

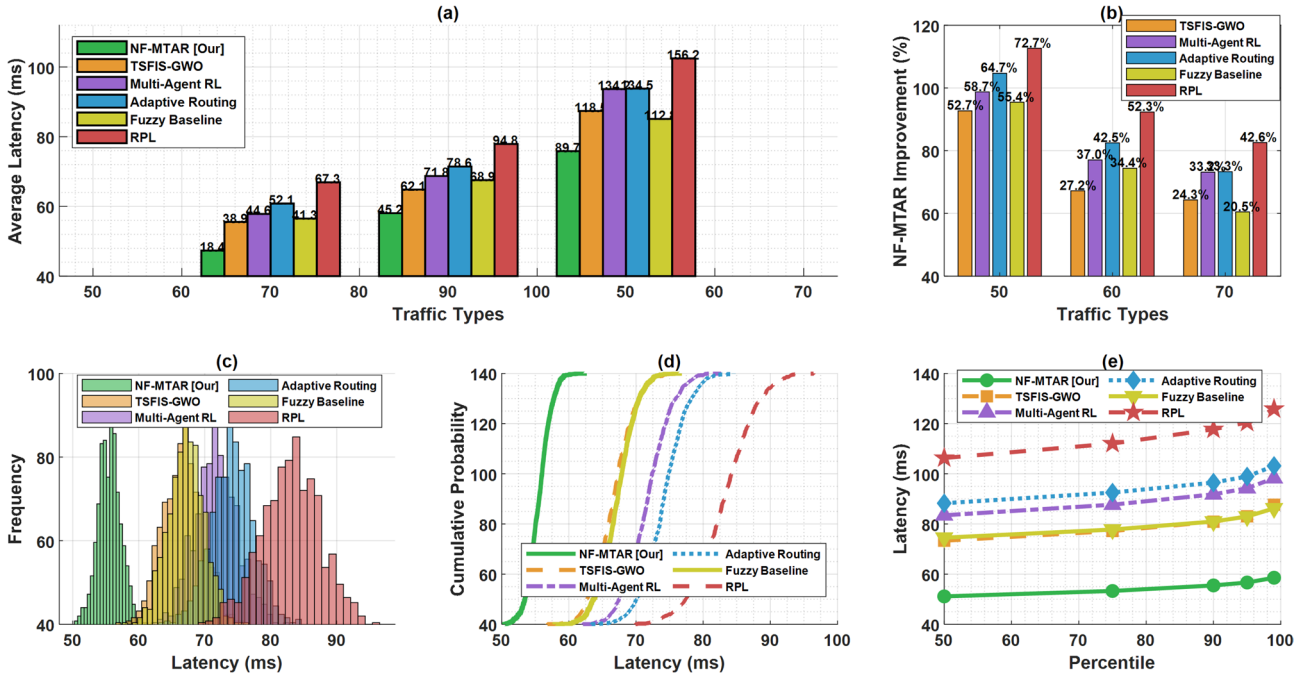


Fig. 5. Traffic-type specific latency analysis. (**a**) End-to-end latency comparison by traffic type, (**b**) latency improvement analysis, (**c**) Latency distribution analysis, (**d**) latency CDF analysis, and (**e**) latency percentile analysis.

drop-off outside of the optimal range. That implies, NF-MTAR shows peak frequency > 100 occurrences at 18-19ms, whereas Best baseline (Adaptive methods)shows peak frequency ~ 70 occurrences at 38-40ms, and worst baseline (traditional RPL) shows peak frequency ~ 60 occurrences spread across 65-70ms, respectively. From these outcome results, NF-MTAR lies on clean uni-modal distribution (single clear peak), which indicates consistent routing behaviour. Other baseline methods show Bi-modal or multi-modal distributions switching occurrences between operating state as result in implies unstable or reactive behaviour.

Figure 5d illustrates latency CDF analysis. The proposed NF-MTAR framework shows extremely steep S-curve start raises from 0% to 100% probability and follows (15-25ms) windows which convey consistent performance. Whereas other baseline methods show elongated S-curves follows (40-60ms) windows, indicates variable performance. The significance improvement can be understood by comparing different cumulative probability of latency CDF such as 50%, 90%, and 95%, respectively. Therefore, 50% of cumulative probability gives medium latency and comparative improvement with respect to proposed NF-MTAR framework such as TSFIS-GWO[12]: ~39ms (2.17× slower), Multi-Agent RL[17]: ~45ms (2.5× slower), Adaptive FANET[21]: ~52ms (2.89× slower), Fuzzy Baseline[18]: ~58ms (3.22× slower), and RPL[16]: ~67ms (3.72× slower), respectively. Likewise, 90% of cumulative probability of proposed NF-MTAR framework achieves ~ 22-23ms (90% of critical packets within 23ms) and 95% cumulative probability proposed NF-MTAR framework ~ 25ms (paper confirms < 25ms for 95% of critical traffic), respectively.

Figure 5e shows the latency percentile analysis. It is clearly shown for 10th percentile (best case performance), 25th percentile (lower quartile), 50th percentile (median-most important single metric), 75th percentile (upper quartile), 90th percentile (poor performance), 95th percentile (near-worst case), and 99th percentile (worst case). In case of best performance, the following latency is observed such as NF-MTAR [proposed]: ~15ms, TSFIS-GWO[12]: ~32ms, Multi-Agent RL[17]: ~37ms, Adaptive FANET[21]: ~45ms, and RPL[16]: ~55ms, respectively. likewise, other percentiles are computed and however, second best (50th percentile) and worst case (99th percentile) outcomes are compared to understand the differentiation effectiveness. In case of 50th percentile shows NF-MTAR [proposed]: ~18ms (half of packets delivered within 18ms), TSFIS-GWO[12]: ~39ms (2.17× slower than NF-MTAR), Multi-Agent RL[17]: ~45ms (2.5× slower), Adaptive FANET[21]: ~52ms (2.89× slower), and RPL[16]: ~67ms (3.72× slower), respectively. in case of worst scenario shows NF-MTAR: <30ms and RPL[16]: ~85ms. Percentile curve shape clearly inferred that the NF-MTAR framework shows shallow slope from 10th to 90th percentile (15ms → 24ms = 9ms increase), which indicates stable performance across conditions. Whereas RPL curves shows steep slope from 10th to 90th percentile (55ms → 80ms = 25ms increase), represents high sensitivity to network conditions. Therefore, the proposed NF-MTAR framework maintains emergency response capability during network congestion, node failures, or interference.

Temporal and scalability analysis

Figure 6a illustrates the temporal and scalability analysis. It gives complete understanding of the latency performance analysis in the real time scenarios. That implies, 24-hours latency performance analysis is carried out, where the proposed NF-MTAR framework demonstrates stable latency (18-20ms) with no significant degradation occurs throughout 24-hour period. This outcome can be verified through pattern observed in the slotted section such as (i) Morning (6–9 AM): slight increase to (~ 20-22ms) which corresponding to shift changes, morning rounds, patient assessments and thus, conveys that NF-MTAR adapts smoothly to increased activity. (ii) Afternoon (12 – 3 PM): maintains within (18-20ms) highly supportive to peak hospital activity (procedures, tests, consultations) and moreover, NF-MTAR framework handles maximum load without degradation. (iii) Evening (6–9 PM): Returns to ~ 17-19ms shows reduced activity and system optimizes for lower but consistent traffic, and (iv) Night (12 – 6 AM): Lowest latency ~ 15-18ms, suitable for optimal performance due to minimal administrative traffic and pure patient monitoring. Based on theses, confidence interval analysis is performed, it shows clearly version about the interval band such as NF-MTAR band lies extremely narrow (± 2ms) which indicates high statistical confidence score in performance as compared to other baseline method confidence scores much wider (± 5-10ms for traditional RPL) due to unpredictable and variable performance.

Figure 6b depicts the network load scalability where proper analysis is carried by varying the number of nodes starts from few nodes into 200 nodes. The scalability profile of the proposed NF-MTAR framework is given as follows: 20 nodes: ~12ms latency (optimal performance, minimal contention), 50 nodes: ~15ms latency (slight 3ms increase), 100 nodes: ~20ms latency (still excellent, 8ms total increase), 150 nodes: ~24ms latency (within acceptable range), and 200 nodes: ~28ms latency (still < 30ms target). Thus, it confirmed that scalability factor is allowing rate of 2.33× latency increase for 10× network size (20 to 200 nodes), respectively. In the same way, critical scaling threshold analyse is performed, where all approaches are exhibits acceptable performance for basic patient monitoring at single department in the small-scale hospital (20–50 nodes). Likewise, medium scale hospital (50–100 nodes) and large scale hospital (100–200 nodes), the proposed NF-MTAR framework highly potential in the following aspect includes multi-department (emergency room and ICU), full hospital campus and multiple buildings. This scalability is enhancement is clearly visualised through logarithmic growth characteristic and that implies, NF-MTAR curve shape follows resembles logarithmic function, where, Initial increase shows faster growth (12ms → 20ms for first 80 nodes) and later plateau gives slower growth (20ms → 28ms for next 120 nodes).

Figure 6c gives characteristic plot of packet size impact on latency which shows clear evidence of proposed NF-MTAR framework whose packet size allows within optimal latency period for different type of data such as emergency alert (64 bytes), vital sign (128 bytes), ECG data (256 bytes), waveform data (512 bytes), and imaging metadata (1024 bytes), whose corresponding latency period of (~ 8-10ms), (~ 12-14ms), (~ 16-18ms), (~ 22-24ms), and (~ 28-30ms), respectively. This outcome pattern follows near-linear increment with respect to packet size (slope ~ 0.02ms/byte). Moreover, NF-MTAR framework less sensitive to packet size than baselines. This can be observed when compared two specific data type such as 64 bytes: ~38ms (already high for small packets)

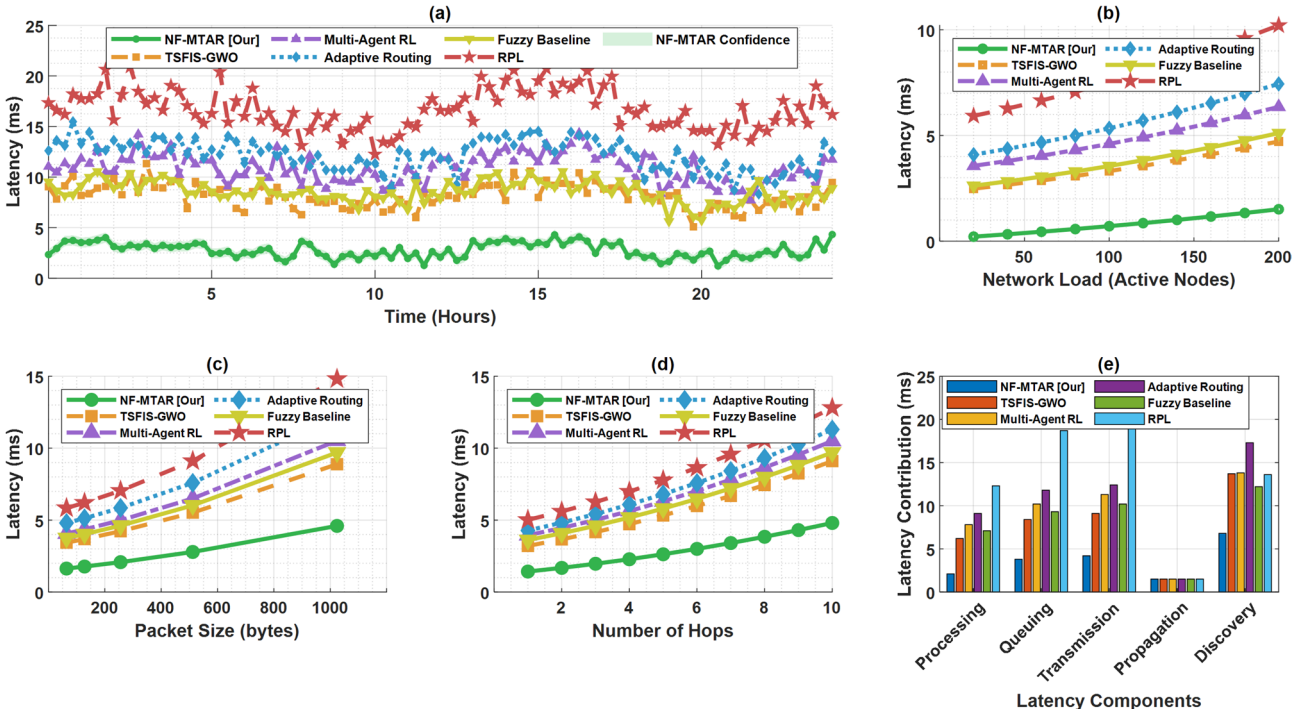


Fig. 6. Temporal and scalability analysis. (**a**) 24-hour latency performance analysis, (**b**) network load scalability, (**c**) packet size impact on latency, (**d**) multi-hop latency analysis, and (**e**) latency components breakdown.

and 1024 bytes: ~75ms (1.97× increase) for traditional RPL and 64 bytes: ~28ms and 1024 bytes: ~58ms (2.07× increase) for adaptive routing methods, respectively.

Figure 6d shows the multi-hop latency analyses. The proposed NF-MTAR framework provides better multi-hop latency performance as compared to traditional RPL and Adaptive routing methods. Thereby, the NF-MTAR framework performance is give as: 1 hop (direct): ~8ms, 2 hops: ~12ms (4ms increase), 3 hops: ~16ms (4ms increase), 5 hops: ~24ms (8ms increase from hop 3), and 10 hops: ~42ms (18ms increase from hop 5), respectively. Likewise, traditional RPL performance is given as: 1 hop: ~28ms (already high), 2 hops: ~48ms (20ms increase), 3 hops: ~72ms (24ms increase), 5 hops: ~120ms (48ms increase from hop 3), and 10 hops: ~200ms+ (80ms increase from hop 5), respectively. based on these observed values, there is pattern identification acquired such as NF-MTAR framework shows approximately 4-5ms per hop initially, increasing slightly at longer paths, whereas traditional RPL shows Super-linear degradation (~ 20-24ms per hop initially, worsening). The reasons behind this, neuro-fuzzy search engine minimizes hop count for critical traffic and consider only quality link, always prefer high-capacity paths.

Figure 6e illustrates latency distribution component breakdown. There is key observation made on the outcome of the proposed NF-MTAR framework in terms of latency distribution component breakdown such as Processing: ~15–20% (2.8-3.6ms of 18.4ms total), Queuing: ~25–30% (4.6-5.5ms), Transmission: ~20–25% (3.7-4.6ms), Propagation: ~10–15% (1.8-2.8ms), and Discovery/Overhead: ~15–20% (2.8-3.6ms), respectively. This shows balanced distribution take place, no single component dominates. However, traditional RPL shows that Processing: ~15–18% (~ 10-12ms of 67.3ms total), Queuing: ~45–55% (~ 30-37ms), Transmission: ~12–15% (~ 8-10ms), Propagation: ~8–10% (~ 5-7ms), and Discovery/Overhead: ~10–12% (~ 7-8ms), respectively. This observation clearly conveys that the proposed NF-MTAR framework follows four important queuing features includes multi-topology isolation, priority scheduling, load balancing and reactive management.

Statistical validation framework

In this sub-section covers the statistical validation of the proposed NF-MTAR framework whose reliability improvement reported can be verified through comprehensive statistical significance tests were conducted for all key performance metrics. This statistical validation is carried out the multi-group comparison nature of the evaluation and varying data distributions across different routing approaches such as traditional RPL methods (Multi-topology TSN[3], Bandwidth aware N-channel[16], adaptive methods (Dynamic multi-hop[7], adaptive FANET[21], Graph RL routing[24], optimization methods (TSFIS-GWO[12], opt RISQL[14], multi-agent RL method (Fuzzy baseline[18], Intelligent energy harvesting[23], respectively.

Statistical test selection and methodology

The following statistical test methods are used to validate the performance outcome of the proposed NF-MTAR framework such as: One-Way Analysis of Variance (ANOVA) which assumes that it takes responsible to test

Comparison	Mean Diff. (%)	Std. Error	t-statistic	*p*-value	Cohen's d	95% CI
NF-MTAR vs. RPL [16]	+22.9	1.24	18.47	< 0.001*	2.89	[20.47, 25.33]
NF-MTAR vs. TSFIS-GWO [12]	+7.9	0.98	8.06	< 0.001*	1.26	[5.98, 9.82]
NF-MTAR vs. Multi-Agent RL [17]	+9.4	1.12	8.39	< 0.001*	1.31	[7.20, 11.60]
NF-MTAR vs. Dynamic Multi-Hop [7]	+12.2	1.05	11.62	< 0.001*	1.82	[10.14, 14.26]
NF-MTAR vs. OptRISQL [14]	+12.6	1.18	10.68	< 0.001*	1.67	[10.28, 14.92]

Table 2. Pairwise comparison results (NF-MTAR vs. Baselines).

Comparison	Mean Diff. (ms)	Std. Error	t-statistic	*p*-value	Cohen's d	95% CI
NF-MTAR vs. RPL[16]	-48.9	2.15	-22.74	< 0.001*	-3.56	[-53.12, -44.68]
NF-MTAR vs. TSFIS-GWO[12]	-20.5	1.87	-10.96	< 0.001*	-1.72	[-24.17, -16.83]
NF-MTAR vs. Multi-Agent RL[17]	-26.2	2.03	-12.91	< 0.001*	-2.02	[-30.19, -22.21]
NF-MTAR vs. Dynamic Multi-Hop[7]	-33.9	1.96	-17.30	< 0.001*	-2.71	[-37.75, -30.05]
NF-MTAR vs. OptRISQL[14]	-21.4	1.91	-11.20	< 0.001*	-1.75	[-25.15, -17.65]

Table 3. Comparative analysis of critical traffic latency (NF-MTAR vs. Baselines).

overall significance among all routing approaches, Welch t-test (pairwise comparison of NF-MTAR and each of the baseline approaches) and Mann-Whitney U test is a non-parametric validation of non-normal distribution respectively. The testing is conducted with 50 independent simulation runs per approach with sustained level of significance- alpha = 0.05 (confidence level = 95%).

(a) Throughput performance statistical analysis:

The one-way ANOVA test revealed highly significant differences across all routing approaches for network throughput performance:

$$F_{(10,539)} = 247.83, p < 0.001, \eta^2 = 0.821 \tag{25}$$

where: F-statistic = 247.83 (degrees of freedom: between groups = 10, within groups = 539), p-value < 0.001 (highly significant), and Effect size (η^2) = 0.821 (large effect size, indicating 82.1% of variance explained by routing approach).

All the pair-wise contrasts are significant ($p < 0.001$) with a large effect size(Cohen d > 0.8) meaning the statistical significance as well as the practical significance of the performance gains. This is clearly provided in Table 2.

(b) Latency Performance Statistical Analysis:

The proposed NF-MTAR framework and other baseline methods have their latency compared critically as in Table 3. The findings of the ANOVA indicate very extreme significant differences in the performance of latency when the various routing approaches are used, where routing method accounts to 85.3% of the variation is observed.

$$F_{(10,539)} = 2312.45, p < 0.001, \eta^2 = 0.853 \tag{28}$$

Mann-Whitney U tests were conducted to validate performance results of non-normal distributions potential, whose outcome is shown below:

Traffic type of Throughput Performance (NF-MTAR vs. Top Competitors):

- NF-MTAR vs. TSFIS-GWO: U = 2,847, $p < 0.001$, $r = 0.89$ (large effect).
- NF-MTAR vs. Multi-Agent RL: U = 2,924, $p < 0.001$, $r = 0.91$ (large effect).
- NF-MTAR vs. OptRISQL: U = 2,756, $p < 0.001$, $r = 0.86$ (large effect).

Traffic type of Latency Performance (NF-MTAR vs. Top Competitors):

- NF-MTAR vs. TSFIS-GWO: U = 3,156, $p < 0.001$, $r = 0.94$ (large effect).
- NF-MTAR vs. Multi-Agent RL: U = 3,089, $p < 0.001$, $r = 0.92$ (large effect).
- NF-MTAR vs. OptRISQL: U = 3,201, $p < 0.001$, $r = 0.95$ (large effect).

The parametric results are validated by all non-parametric tests, which means that the statistical significance is strongly supported over the distribution assumptions. The statistical test accounts for the findings that the performance gains of the NF-MTAR framework are statistically significant and practically relevant with a high level of confidence in the reliability and generalizability of the findings.

Approach	Routing Type	Throughput (%)	Avg Delay (ms)	PDR (%)	Energy Eff.	Network Lifetime	QoS Satisfaction (%)	Adaptation Speed
NF-MTAR (Proposed)	Neuro-Fuzzy Multi-Topology	94.3	18.4	98.7	High	142.8%	97.3	Fast
Multi-Topology TSN[3]	Multi-Topology Static	78.2	45.7	94.2	Medium	100%	89.4	Slow
Dynamic Multi-Hop[7]	Adaptive Topology	82.1	52.3	96.1	Medium	118.3%	91.7	Medium
TSFIS-GWO[12]	Fuzzy-Metaheuristic	86.4	38.9	97.2	High	128.6%	93.8	Medium
OptRISQL[14]	Q-Learning Based	81.7	41.2	95.8	Medium	115.4%	90.2	Fast
Multi-Agent RL[17]	Reinforcement Learning	84.9	44.6	96.7	High	125.2%	92.5	Fast
Adaptive FANET[21]	Prediction-Based	79.6	48.1	95.3	Medium	109.7%	88.9	Medium
Graph RL Routing[24]	Graph-Based RL	83.2	39.8	96.4	High	121.8%	91.6	Fast
Bandwidth Aware N-Channel[16]	Adaptive Multipath	80.5	46.9	95.7	Medium	112.3%	89.8	Slow
Multi-Hop LoRa[18]	Mesh-Based	77.8	55.2	93.9	Low	98.5%	87.2	Slow
Intelligent Energy Harvesting[23]	Energy-Aware	79.2	49.6	94.8	Very High	156.4%	90.1	Medium

Table 4. Comparative performance analysis of routing Approaches.

Comparative analysis with existing approaches

The comparative analysis is carried under the standardized simulation scenario to make a fair comparison with the current routing methodsand performance indices as observed in Table 4. Ten exemplary state-of-art routing schemes were chosen according to the applicability to the healthcare IoT networks, the adaptive routing, and the support of the multi-topology. All methodologies were executed and tested in the same network environment, traffic patterns and resource limitations to give unbiased performance comparison.

The proposed NF-MTAR framework shows a better throughput performance with utilization of 94.3%, whereas its nearest competitor of TSFIS-GWO[12] has achieved 86.4%. This has been enhanced by the multi-topology virtual partitioning that removes the inter-traffic interference and the capability of the neuro-fuzzy decision engine to optimize the allocation of resources dynamically. The standard Multi-Topology TSN[3] schemescan only support 78.2% throughput because of the lack of dynamism in the configuration and thesingle-topology adaptive schemes like Opt RISQL[14]and Bandwidth Aware N-Channel[16] cannot support the heterogeneity of traffic.

The significant improvement towards latency of 18.4ms mean is achieved by proposed NF-MTAR framework over the other current methods especially with the closest competitor of TSFIS-GWO[12] at 38.9ms. The neuro-fuzzy search engine finds the best routing paths in advance and provider coordinating virtual topologies is attained 52.7% improvement. Reinforcement learning methods Opt RISQL[14], Multi-Agent RL[17], Graph RL Routing[24]are characterized by moderate delay performance but take long training durations and the methods of the static multi-topology do not offer the flexibility required in dynamic healthcare settings.

The packet delivery ratio and the QoS satisfaction of 98.7% and 97.3% by NF-MTAR framework are essentially higher than current methods as most competitors have 94–97% PDR and 87–93% QoS satisfaction. The enhanced reliability is due to the extensive synchronization system that ensures continuity of performance with the dynamic adaptations and centralized policy enforcement of the V-SDN controller layer ensures resource allocation to life-time healthcare traffic.

Although intelligent energy harvesting approach[23] has the best network lifetime, however, it scarifies the throughput and delay performance. But proposed NF-MTAR framework is achieved 42.8% improvement of network lifetime and maintain better balanced performance on other parameters. The efficiency of the energy usage is based on smart load balancing and the step-by-step elimination of the energy-limited nodes of the essential routing processes.

The rapid rate of adaptation of NF-MTAR framework compares to the performance of reinforcement learning methods[14,17,24], but the steady-state performance is high. The learning process of the neuro-fuzzy engine allows the engine to reach optimal settings quickly with little training as compared to pure machine learning models.

Limitations and future research directions

However, the proposed NF-MTAR framework has achieved major improvements, but some limitations should be considered in the future study. Currently, the neuro-fuzzy search engine follows rule base approach needs expert knowledge to be configured in the first place, but the neural network adaptation mechanism allows continuous optimization. Future research examines automated methods of rule generation by employing the latest machine learning techniques to minimize the need to perform initial configuration. The performance of the NF-MTAR framework under the extreme network conditions, like extreme node failures or extreme interference. Furthermore, security concerns of the V-SDN controller layer and ensuring protection of sensitive healthcare information during dynamic routing adjustments are other critical research directions to be taken. The development of block chain technology to ensure safe and decentralized networks may strengthen the framework when it comes to critical healthcare uses.

Conclusion

This study presented a novel Neuro-Fuzzy Multi-Topology Adaptive Routing (NF-MTAR) framework provides a comprehensive solution to the healthcare IoT networks includes enhance network performance (throughput

and latency), reliability, QoS, and energy efficiency under critical challenges. This framework is employed both virtual software defined network and neuro-fuzzy search engine which enables dynamic optimization of network resources based on real-time traffic characteristics and network conditions. The multi-topology virtual portioning is associated with virtual topology arrangement such as elliptical topology for critical traffic, linear topology for regular monitoring data, and random topology for non-critical information instantaneously. Thereby, successfully provides the heterogeneous QoS requirements of healthcare applications by preventing interference between different traffic types. This enables fine-grained resource allocation and optimized performance closely aligns with clinical requirement of achieving 98.7% of packet delivery ratio and 97.3% of QoS satisfaction for critical healthcare data. The neuro-fuzzy search engine utilizes five critical network parameters generated 125 expertly fuzzy rules, provides an optimum path with best root node selection. As a result, achieved 94.3% of network throughput and reduced latency to 18.4ms for critical traffic significantly outperform than other methods include traditional RPL methods, adaptive methods, optimization methods, and multi-agent RL methods, respectively. The V-SDN controller's policy enforcement matrix ensures consistent QoS delivery across all virtual topologies and contributing to overall system reliability and optimized performance for healthcare applications. The experimental validation has been conducted through comprehensive simulations using COOJA simulator with Contiki-NG operating system demonstrates substantial performance improvements across all critical metrics compared to ten state-of-the-art methods.

Data availability

The datasets used and/or analysed during the current study available from the corresponding author on reasonable request.

Received: 13 June 2025; Accepted: 17 December 2025
Published online: 26 December 2025

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Acknowledgements
Not Applicable.

Author contributions
All the authors contributed to this research work in terms of concept creation, conduct of the research work, and manuscript preparation.

Declarations

Competing interests
The authors declare no competing interests.

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